Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Gerrit Hofman Dairy

Physical address of dairy:

4820 Perrin RD	Manteca	San Joaquin	95337
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 09/01/1972

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X257-X030-XX21-XXXX

B. OPERATORS

Hofman, Jim

Operator name: Hofman, Jim

Telephone no.: (209) 823-3000 Landline, (209) 612-3064 Cellular

4820 Perrin RD	Manteca	CA	95337
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Hofman, Jim

Legal owner name: Hofman, Jim

Telephone no.: (209) 823-3000 Landline, (209) 612-3064 Cellular

4820 Perrin RD	Manteca	CA	95337
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	250	250	250
Number under roof	1,000	120	300	0	0	0
Maximum number	1,000	120	300	250	250	250
Average number	1,000	120	300	250	250	250
Avg live weight (lbs)	1,400	1,450	1,000	700		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 34,877.75 *tons per reporting period*

Total nitrogen from manure: 438,094.02 *lbs per reporting period*

After ammonia losses (30% loss applied): 306,665.81 *lbs per reporting period*

Total phosphorus from manure: 70,872.67 *lbs per reporting period*

Total potassium from manure: 192,823.23 *lbs per reporting period*

Total salt from manure: 498,444.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 21,394,500 *gallons*

Total nitrogen generated: 98,789.75 *lbs*

Total phosphorus generated: 11,426.65 *lbs*

Total potassium generated: 171,779.93 *lbs*

Total salt generated: 964,849.40 *lbs*

	21,394,500	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	21,394,500	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Ground water
Stanislaus Water Users	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	Corral solids	1,760.00 *ton*	Dry-weight	55.9		24,200.00	10,400.00	48,800.00		28.30
10/31/2023	Corral solids	1,720.00 *ton*	Dry-weight	32.6		23,200.00	9,560.00	54,600.00		38.60

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	91,356.74	38,309.56	202,346.59	1,334,270.72
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	91,356.74	38,309.56	202,346.59	1,334,270.72

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	31	31	2	process wastewater	X257-X030-XX21-XXXX
2	21	21	2	process wastewater	X257-X030-XX21-XXXX
3	35	35	2	process wastewater	X257-X030-XX21-XXXX
4	32	32	2	process wastewater	X257-X030-XX21-XXXX
5	40	40	2	process wastewater	X257-X030-XX21-XXXX
6	39	39	2	process wastewater	X257-X030-XX05-XXXX X257-X030-XX21-XXXX
Totals for areas that were used for application	198	198	12		
Totals for areas that were not used for application					
Land application area totals	198	198	12		

B. CROPS AND HARVESTS

1

Field name: 1

11/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 31 Plant date: 11/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	580.00 *ton*	As-is		71.3	5,400.00	1,090.00	10,900.00		12.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	18.71	202.06	40.79	407.87	1,342.42

1

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 31 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	875.00 *ton*	As-is		62.6	3,930.00	1,250.00	4,800.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.23	221.85	70.56	270.97	1,076.76

2

Field name: 2

11/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 11/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/07/2023	426.00 *ton*	As-is		71.0	5,710.00	1,260.00	11,400.00		12.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	20.29	231.66	51.12	462.51	1,506.01

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	665.00 *ton*	As-is		68.4	3,950.00	750.00	5,270.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.67	250.17	47.50	333.77	1,160.77

3

Field name: 3

11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	700.00 *ton*	As-is		66.8	3,920.00	910.00	6,160.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	20.00	156.80	36.40	246.40	1,142.08

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	1,050.00 *ton*	As-is		61.8	3,630.00	790.00	4,030.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	217.80	47.40	241.80	1,306.44

4

Field name: 4

11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 32 Plant date: 11/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	630.00 *ton*	As-is		70.8	5,640.00	1,170.00	11,500.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	19.69	222.08	46.07	452.81	1,414.19

4

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 32 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	982.00 *ton*	As-is		67.8	3,410.00	820.00	4,850.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.69	209.29	50.33	297.67	1,403.16

5

Field name: 5

11/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	815.00 *ton*	As-is		71.4	6,090.00	1,170.00	12,100.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	20.38	248.17	47.68	493.08	1,433.50

05/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	1,391.00 *ton*	As-is		71.2	3,310.00	1,130.00	6,980.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	34.78	230.21	78.59	485.46	1,722.61

6

Field name: 6

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 39 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	800.00 *ton*	As-is		63.1	3,870.00	1,030.00	7,480.00		9.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	20.51	158.77	42.26	306.87	1,438.15

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	1,305.00 *ton*	As-is		70.4	3,580.00	1,100.00	7,220.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	33.46	239.58	73.62	483.18	1,584.74

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/22/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 11/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.63	18.64	267.16	795.73	357,000.00 *gal*
Canal	Ground water	2.06	0.00	0.00	7.35	2,730,000.00 *gal*
Application event totals		82.69	18.64	267.16	803.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	133.86	21.95	321.26	1,256.47	663,000.00 *gal*
Application event totals		133.86	21.95	321.26	1,256.47	

1 - 05/25/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	105.12	12.43	211.29	923.40	484,500.00 *gal*
Canal	Ground water	0.00	0.00	0.00	35.91	3,705,000.00 *gal*
Application event totals		105.12	12.43	211.29	959.31	

1 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	39.68	4,095,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.23	1.78	24.84	476.94	459,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	34.02	3,510,000.00 *gal*
Application event totals		44.23	1.78	24.84	510.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.23	1.78	24.84	476.94	459,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	34.02	3,510,000.00 *gal*
Application event totals		44.23	1.78	24.84	510.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	37.79	3,900,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.78	1.68	23.46	450.44	433,500.00 *gal*
Canal	Ground water	0.00	0.00	0.00	32.13	3,315,000.00 *gal*
Application event totals		41.78	1.68	23.46	482.57	

1 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.78	1.68	23.46	450.44	433,500.00 *gal*
Canal	Ground water	0.00	0.00	0.00	32.13	3,315,000.00 *gal*
Application event totals		41.78	1.68	23.46	482.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	35.91	3,705,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.91	

2 - 11/22/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough Plant date: 11/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.20	14.96	218.88	856.05	306,000.00 *gal*
Application event totals		91.20	14.96	218.88	856.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	136.80	22.43	328.32	1,284.08	459,000.00 *gal*
Application event totals		136.80	22.43	328.32	1,284.08	

2 - 05/25/2023: Corn, silage

Field name: 2

Crop: Corn, silage Plant date: 05/25/2023

2 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	106.18	12.55	213.41	932.66	331,500.00 *gal*
Canal	Ground water	0.00	0.00	0.00	36.26	2,535,000.00 *gal*
Application event totals		106.18	12.55	213.41	968.93	

06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	39.05	2,730,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.05	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.53	1.75	24.44	469.37	306,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	33.48	2,340,000.00 *gal*
Application event totals		43.53	1.75	24.44	502.85	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.53	1.75	24.44	469.37	306,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	33.48	2,340,000.00 *gal*
Application event totals		43.53	1.75	24.44	502.85	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	39.05	2,730,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.05	

2 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.53	1.75	24.44	469.37	306,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	33.48	2,340,000.00 *gal*
Application event totals		43.53	1.75	24.44	502.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.53	1.75	24.44	469.37	306,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	33.48	2,340,000.00 *gal*
Application event totals		43.53	1.75	24.44	502.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	36.26	2,535,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.26	

3 - 11/23/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 11/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	96.92	22.41	321.14	956.49	484,500.00 *gal*
Canal	Ground water	2.34	0.00	0.00	8.37	3,510,000.00 *gal*
Application event totals		99.26	22.41	321.14	964.86	

3 - 11/23/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	104.88	17.20	251.71	984.46	586,500.00 *gal*
Application event totals		104.88	17.20	251.71	984.46	

3 - 05/26/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	107.81	12.75	216.69	947.01	561,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	36.82	4,290,000.00 *gal*
Application event totals		107.81	12.75	216.69	983.83	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	40.17	4,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.17	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.53	1.75	24.44	469.37	510,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	33.48	3,900,000.00 *gal*
Application event totals		43.53	1.75	24.44	502.85	

3 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.36	1.66	23.22	445.90	484,500.00 *gal*
Canal	Ground water	0.00	0.00	0.00	31.80	3,705,000.00 *gal*
Application event totals		41.36	1.66	23.22	477.70	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	36.82	4,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.82	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.36	1.66	23.22	445.90	484,500.00 *gal*
Canal	Ground water	0.00	0.00	0.00	31.80	3,705,000.00 *gal*
Application event totals		41.36	1.66	23.22	477.70	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.36	1.66	23.22	445.90	484,500.00 *gal*
Canal	Ground water	0.00	0.00	0.00	31.80	3,705,000.00 *gal*
Application event totals		41.36	1.66	23.22	477.70	

09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	35.15	4,095,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.15	

4 - 11/23/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough

Plant date: 11/23/2022

4 - 11/23/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	104.74	17.18	251.37	983.12	535,500.00 *gal*
Application event totals		104.74	17.18	251.37	983.12	

03/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	119.70	19.63	287.28	1,123.57	612,000.00 *gal*
Application event totals		119.70	19.63	287.28	1,123.57	

4 - 05/26/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	107.20	12.67	215.46	941.63	510,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	36.61	3,900,000.00 *gal*
Application event totals		107.20	12.67	215.46	978.24	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	40.28	4,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.28	

4 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.23	1.82	25.40	487.71	484,500.00 *gal*
Canal	Ground water	0.00	0.00	0.00	34.78	3,705,000.00 *gal*
Application event totals		45.23	1.82	25.40	522.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.85	1.72	24.06	462.04	459,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	32.95	3,510,000.00 *gal*
Application event totals		42.85	1.72	24.06	494.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	38.44	4,095,000.00 *gal*
Application event totals		0.00	0.00	0.00	38.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.85	1.72	24.06	462.04	459,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	32.95	3,510,000.00 *gal*
Application event totals		42.85	1.72	24.06	494.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	32.95	3,510,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.95	

4 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.85	1.72	24.06	462.04	459,000.00 *gal*
Application event totals		42.85	1.72	24.06	462.04	

09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	36.61	3,900,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.61	

5 - 11/25/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough Plant date: 11/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	119.70	19.63	287.28	1,123.57	765,000.00 *gal*
Application event totals		119.70	19.63	287.28	1,123.57	

03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	123.69	20.28	296.85	1,161.02	790,500.00 *gal*
Application event totals		123.69	20.28	296.85	1,161.02	

5 - 05/27/2023: Corn, silage

Field name: 5

Crop: Corn, silage Plant date: 05/27/2023

5 - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	107.20	12.67	215.46	941.63	637,500.00 *gal*
Canal	Ground water	0.10	0.00	0.00	36.61	4,875,000.00 *gal*
Application event totals		107.30	12.67	215.46	978.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.11	0.00	0.00	39.54	5,265,000.00 *gal*
Application event totals		0.11	0.00	0.00	39.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.80	1.76	24.59	472.30	586,500.00 *gal*
Canal	Ground water	0.09	0.00	0.00	33.68	4,485,000.00 *gal*
Application event totals		43.90	1.76	24.59	505.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.90	1.69	23.52	451.77	561,000.00 *gal*
Canal	Ground water	0.09	0.00	0.00	32.22	4,290,000.00 *gal*
Application event totals		41.99	1.69	23.52	483.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.90	1.69	23.52	451.77	561,000.00 *gal*
Canal	Ground water	0.09	0.00	0.00	32.22	4,290,000.00 *gal*
Application event totals		41.99	1.69	23.52	483.99	

5 - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.90	1.69	23.52	451.77	561,000.00 *gal*
Canal	Ground water	0.09	0.00	0.00	32.22	4,290,000.00 *gal*
Application event totals		41.99	1.69	23.52	483.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.10	0.00	0.00	36.61	4,875,000.00 *gal*
Application event totals		0.10	0.00	0.00	36.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.10	0.00	0.00	35.15	4,680,000.00 *gal*
Application event totals		0.10	0.00	0.00	35.15	

6 - 11/26/2022: Oats, silage-soft dough

Field name: 6

Crop: Oats, silage-soft dough

Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.85	13.42	196.43	768.25	510,000.00 *gal*
Application event totals		81.85	13.42	196.43	768.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	130.95	21.48	314.29	1,229.21	816,000.00 *gal*
Application event totals		130.95	21.48	314.29	1,229.21	

6 - 05/28/2023: Corn, silage

Field name: 6

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	105.55	12.48	212.14	927.14	612,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	36.05	4,680,000.00 *gal*
Application event totals		105.55	12.48	212.14	963.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	39.05	5,070,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.93	1.81	25.22	484.41	586,500.00 *gal*
Canal	Ground water	0.00	0.00	0.00	34.55	4,485,000.00 *gal*
Application event totals		44.93	1.81	25.22	518.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.97	1.73	24.13	463.35	561,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	33.05	4,290,000.00 *gal*
Application event totals		42.97	1.73	24.13	496.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.97	1.73	24.13	463.35	561,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	33.05	4,290,000.00 *gal*
Application event totals		42.97	1.73	24.13	496.40	

6 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.97	1.73	24.13	463.35	561,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	33.05	4,290,000.00 *gal*
Application event totals		42.97	1.73	24.13	496.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	36.05	4,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.05	

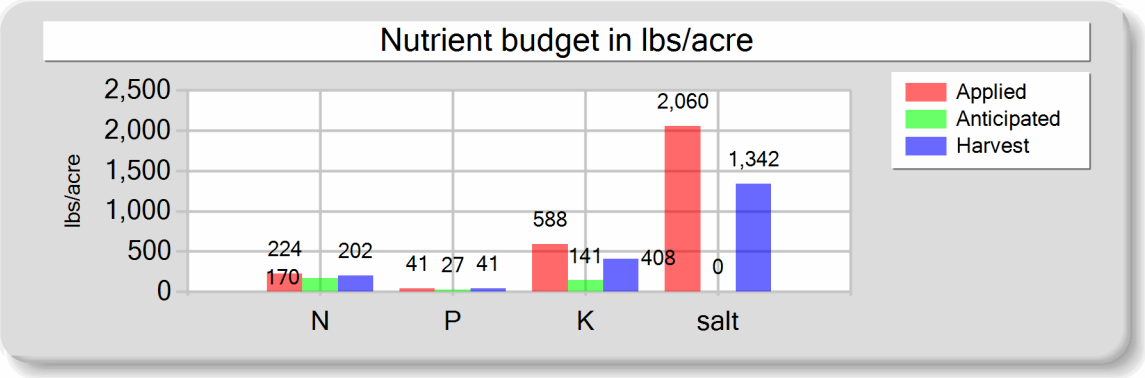
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	34.55	4,485,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.55	

B. NUTRIENT BUDGET

1 - 11/22/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	214.49	40.60	588.42	2,052.19
Fresh water	2.06	0.00	0.00	7.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	223.54	40.60	588.42	2,059.54
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	202.06	40.79	407.87	1,342.42
Nutrient balance	21.48	-0.19	180.55	717.12
Applied to removed ratio	1.11	1.00	1.44	1.53

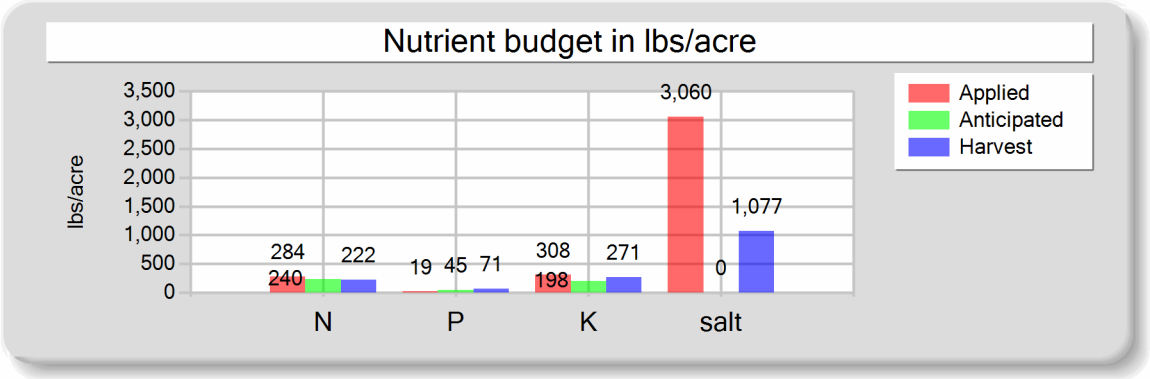
Fresh water applied	
2,730,000.00	*gallons*
100.54	*acre-inches*
3.24	*inches/acre*

Process wastewater applied	
1,020,000.00	*gallons*
37.56	*acre-inches*
1.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 05/25/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	277.14	19.35	307.87	2,778.17
Fresh water	0.00	0.00	0.00	281.57
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	284.14	19.35	307.87	3,059.75
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	221.85	70.56	270.97	1,076.76
Nutrient balance	62.29	-51.22	36.90	1,982.99
Applied to removed ratio	1.28	0.27	1.14	2.84

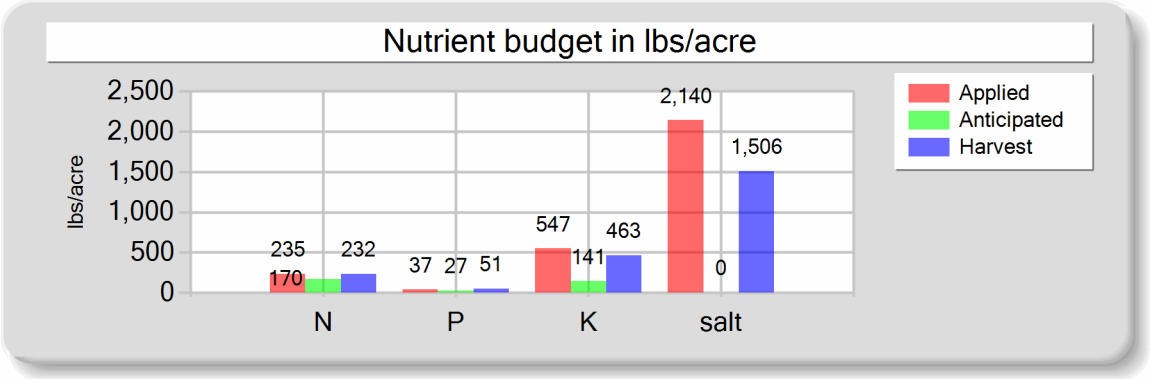
Fresh water applied	
29,055,000.00	*gallons*
1,070.00	*acre-inches*
34.52	*inches/acre*

Process wastewater applied	
2,269,500.00	*gallons*
83.58	*acre-inches*
2.70	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/22/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.00	37.39	547.19	2,140.13
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.00	37.39	547.19	2,140.13
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	231.66	51.12	462.51	1,506.01
Nutrient balance	3.33	-13.73	84.68	634.12
Applied to removed ratio	1.01	0.73	1.18	1.42

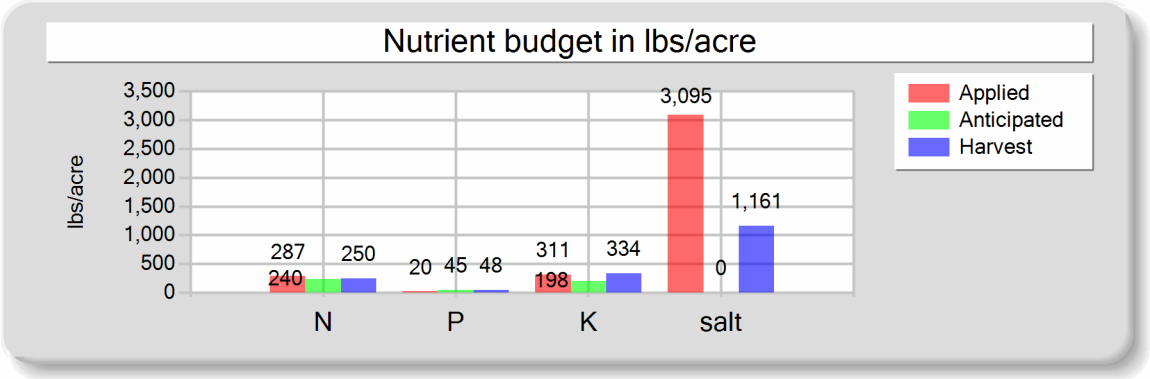
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
765,000.00	*gallons*
28.17	*acre-inches*
1.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 05/25/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	280.30	19.56	311.17	2,810.14
Fresh water	0.00	0.00	0.00	284.54
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	287.30	19.56	311.17	3,094.68
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	250.17	47.50	333.77	1,160.77
Nutrient balance	37.14	-27.94	-22.60	1,933.91
Applied to removed ratio	1.15	0.41	0.93	2.67

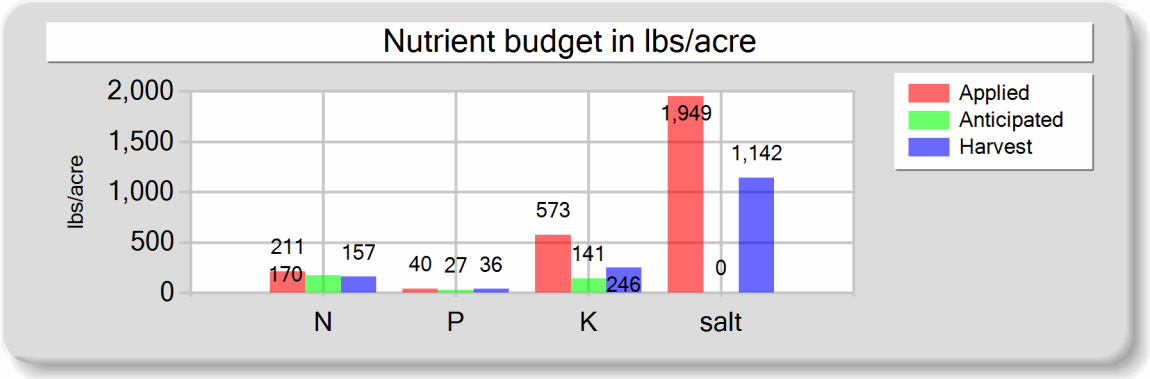
Fresh water applied	
19,890,000.00	*gallons*
732.48	*acre-inches*
34.88	*inches/acre*

Process wastewater applied	
1,555,500.00	*gallons*
57.28	*acre-inches*
2.73	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 11/23/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 11/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.80	39.61	572.85	1,940.96
Fresh water	2.34	0.00	0.00	8.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	211.14	39.61	572.85	1,949.33
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	156.80	36.40	246.40	1,142.08
Nutrient balance	54.34	3.21	326.45	807.25
Applied to removed ratio	1.35	1.09	2.32	1.71

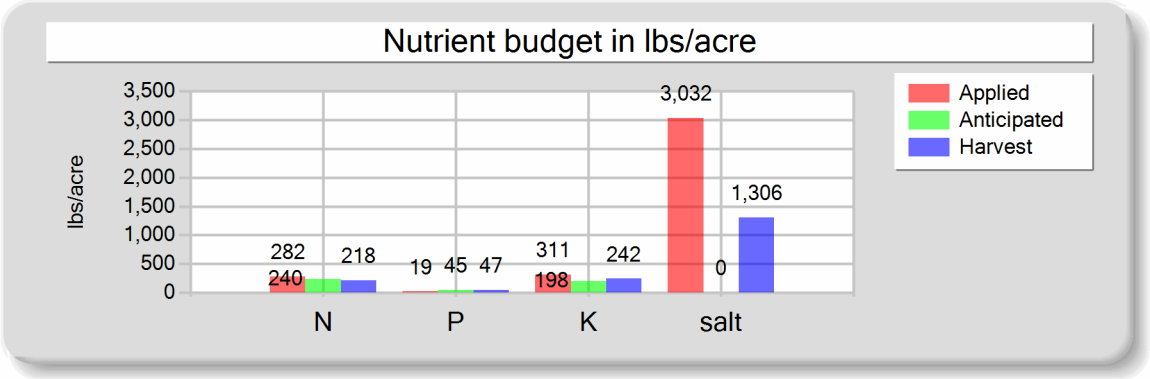
Fresh water applied	
3,510,000.00	*gallons*
129.26	*acre-inches*
3.69	*inches/acre*

Process wastewater applied	
1,071,000.00	*gallons*
39.44	*acre-inches*
1.13	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 05/26/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	275.41	19.49	310.79	2,754.09
Fresh water	0.00	0.00	0.00	277.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.41	19.49	310.79	3,031.93
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	217.80	47.40	241.80	1,306.44
Nutrient balance	64.61	-27.91	68.99	1,725.49
Applied to removed ratio	1.30	0.41	1.29	2.32

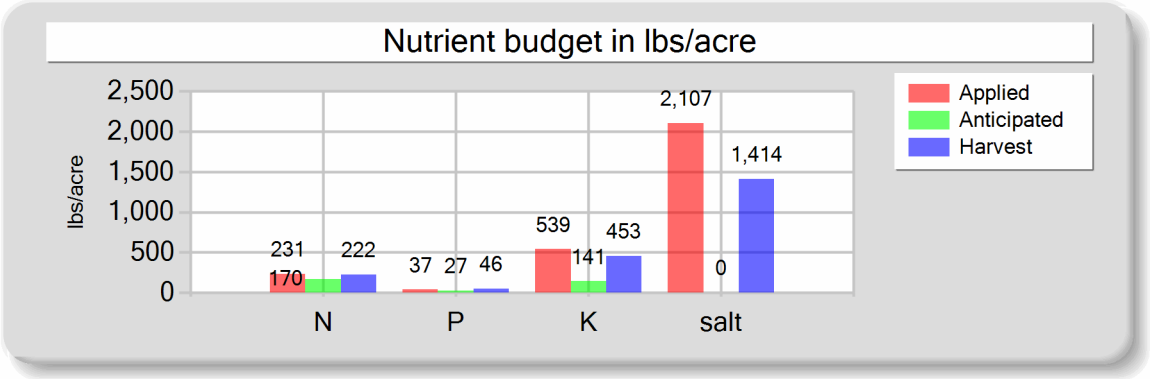
Fresh water applied	
32,370,000.00	*gallons*
1,192.08	*acre-inches*
34.06	*inches/acre*

Process wastewater applied	
2,524,500.00	*gallons*
92.97	*acre-inches*
2.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 11/23/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 11/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	224.43	36.81	538.64	2,106.70
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	231.43	36.81	538.64	2,106.70
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	222.08	46.07	452.81	1,414.19
Nutrient balance	9.36	-9.26	85.83	692.50
Applied to removed ratio	1.04	0.80	1.19	1.49

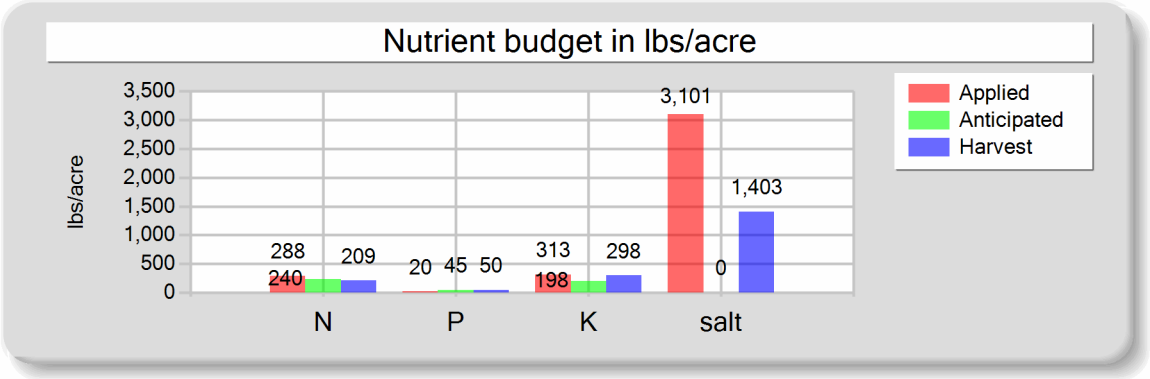
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
1,147,500.00 *gallons*
42.26 *acre-inches*
1.32 *inches/acre*

Total harvests for the crop
1 *harvests*

4 - 05/26/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	280.99	19.67	313.03	2,815.44
Fresh water	0.00	0.00	0.00	285.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	287.99	19.67	313.03	3,101.03
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	209.29	50.33	297.67	1,403.16
Nutrient balance	78.70	-30.66	15.36	1,697.88
Applied to removed ratio	1.38	0.39	1.05	2.21

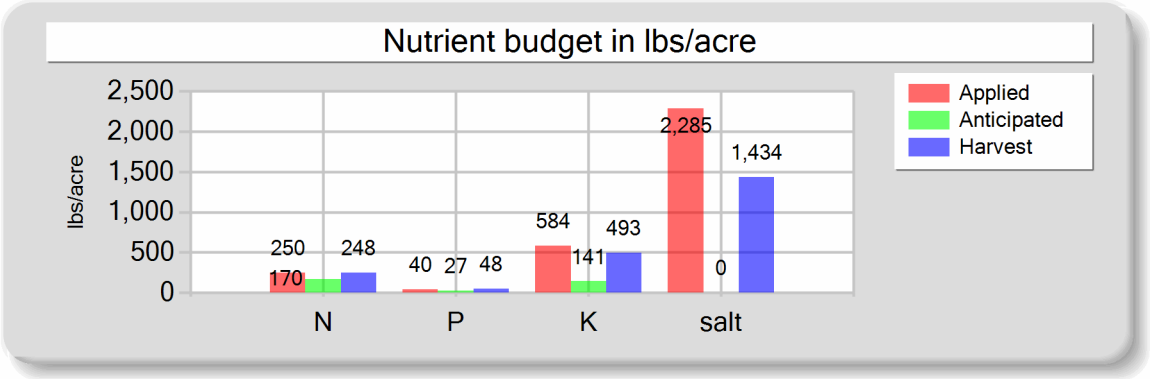
Fresh water applied	
30,420,000.00	*gallons*
1,120.27	*acre-inches*
35.01	*inches/acre*

Process wastewater applied	
2,371,500.00	*gallons*
87.33	*acre-inches*
2.73	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 11/25/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 11/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	243.39	39.92	584.13	2,284.59
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	250.39	39.92	584.13	2,284.59
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	248.17	47.68	493.08	1,433.50
Nutrient balance	2.22	-7.76	91.05	851.09
Applied to removed ratio	1.01	0.84	1.18	1.59

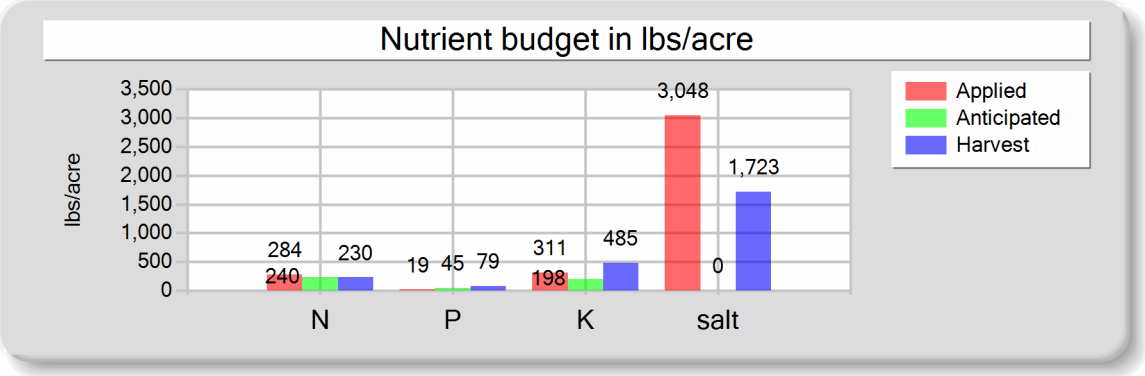
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
1,555,500.00 *gallons*
57.28 *acre-inches*
1.43 *inches/acre*

Total harvests for the crop
1 *harvests*

5 - 05/27/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 05/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	276.70	19.49	310.63	2,769.24
Fresh water	0.77	0.00	0.00	278.26
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	284.47	19.49	310.63	3,047.50
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	230.21	78.59	485.46	1,722.61
Nutrient balance	54.26	-59.10	-174.83	1,324.89
Applied to removed ratio	1.24	0.25	0.64	1.77

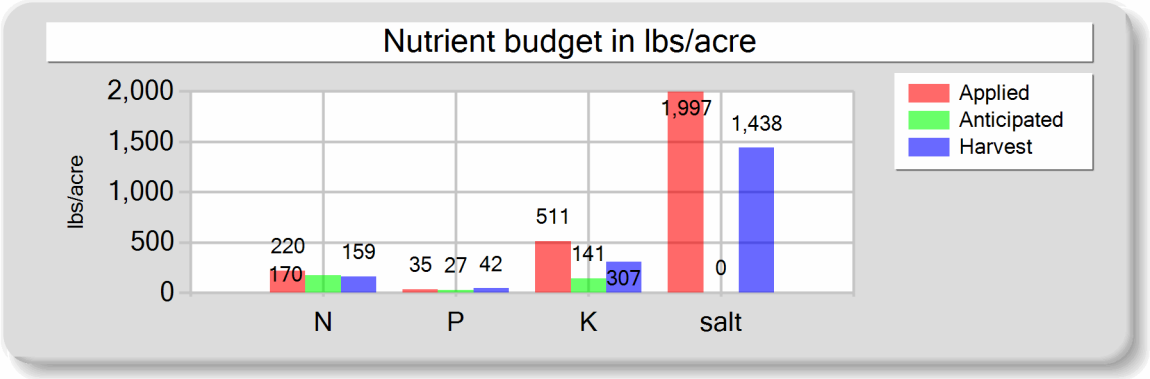
Fresh water applied	
37,050,000.00	*gallons*
1,364.43	*acre-inches*
34.11	*inches/acre*

Process wastewater applied	
2,907,000.00	*gallons*
107.05	*acre-inches*
2.68	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 11/26/2022: Oats, silage-soft dough

Field name: 6 Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	212.80	34.90	510.71	1,997.46
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	219.80	34.90	510.71	1,997.46
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	158.77	42.26	306.87	1,438.15
Nutrient balance	61.03	-7.36	203.84	559.31
Applied to removed ratio	1.38	0.83	1.66	1.39

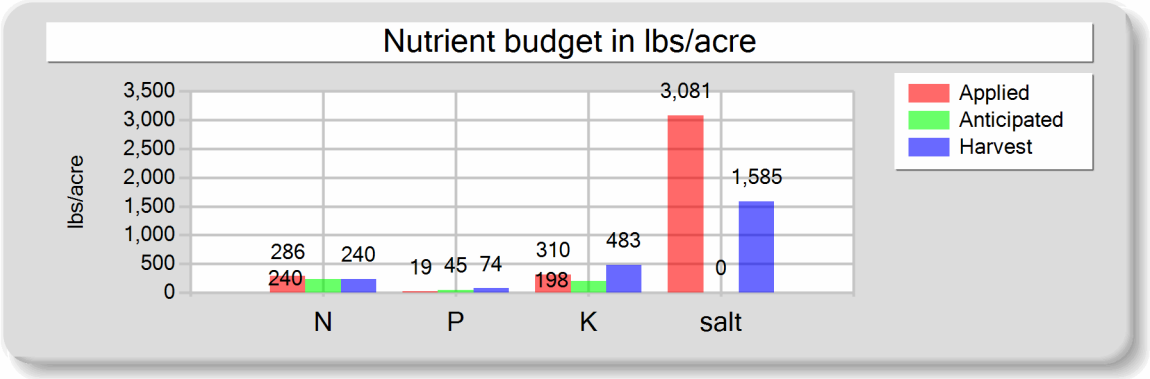
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
1,326,000.00 *gallons*
48.83 *acre-inches*
1.25 *inches/acre*

Total harvests for the crop
1 *harvests*

6 - 05/28/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	279.40	19.47	309.75	2,801.62
Fresh water	0.00	0.00	0.00	279.39
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	286.40	19.47	309.75	3,081.01
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	239.58	73.62	483.18	1,584.74
Nutrient balance	46.81	-54.14	-173.43	1,496.27
Applied to removed ratio	1.20	0.26	0.64	1.94

Fresh water applied	
36,270,000.00	*gallons*
1,335.70	*acre-inches*
34.25	*inches/acre*

Process wastewater applied	
2,881,500.00	*gallons*
106.12	*acre-inches*
2.72	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Export Manure

Sample and source description: Export Manure

Sample date: 05/11/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,200.00	10,400.00	48,800.00							28.30
DL	100.00	100.00	100.00							0.01

Fall Export Manure

Sample and source description: Fall Export Manure

Sample date: 10/31/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 32.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,200.00	9,560.00	54,600.00	10,900.00	6,830.00	4,930.00	5,810.00	907.00		38.60
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 11/22/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	839.00	410.00	0.00	0.00	194.00	2,780.00								10.20	8,280
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 01/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	750.00	364.00	0.00	0.00	123.00	1,800.00								92,400.00	7,040
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 05/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	806.00	398.00	0.00	0.00	95.30	1,620.00								9,620.00	7,080
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 08/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	358.00	230.00	0.00	0.00	14.40	201.00								6,010.00	3,860
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 10/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	476.00	283.00	0.00	0.00	54.80	1,550.00								7,750.00	5,760
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal

Fresh Water

Sample description: Fresh Water

Sample date: 05/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								65.00	36
DL	0.50	0.50	0.50								1.00	10

Stanislaus Water Users

Fresh Water

Sample description: Fresh Water

Sample date: 08/03/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.80	0.00	2.80								100.00	10
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 11/22/2022: Oats, silage-soft dough

1 - 11/22/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,400.00	1,090.00	10,900.00		12.50
DL	100.00	100.00	100.00		0.01

1 - 05/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,930.00	1,250.00	4,800.00		5.10
DL	100.00	100.00	100.00		0.01

2 - 11/22/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,710.00	1,260.00	11,400.00		12.80
DL	100.00	100.00	100.00		0.01

2 - 05/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,950.00	750.00	5,270.00		5.80
DL	100.00	100.00	100.00		0.01

3 - 11/23/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,920.00	910.00	6,160.00		8.60
DL	100.00	100.00	100.00		0.01

3 - 05/26/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,630.00	790.00	4,030.00		5.70
DL	100.00	100.00	100.00		0.01

4 - 11/23/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,640.00	1,170.00	11,500.00		12.30
DL	100.00	100.00	100.00		0.01

4 - 05/26/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,410.00	820.00	4,850.00		7.10
DL	100.00	100.00	100.00		0.01

5 - 11/25/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,090.00	1,170.00	12,100.00		12.30
DL	100.00	100.00	100.00		0.01

5 - 05/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,310.00	1,130.00	6,980.00		8.60
DL	100.00	100.00	100.00		0.01

6 - 11/26/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,870.00	1,030.00	7,480.00		9.50
DL	100.00	100.00	100.00		0.01

6 - 05/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,580.00	1,100.00	7,220.00		8.00
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

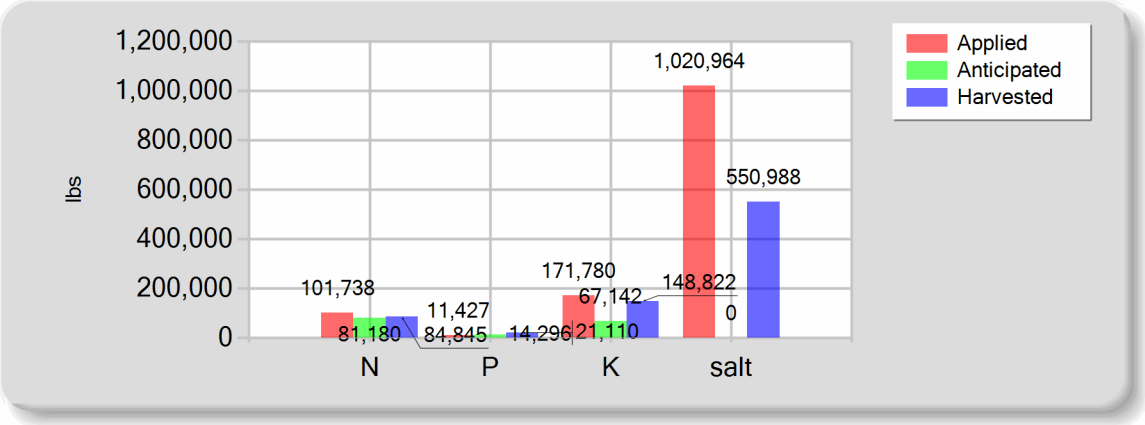
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

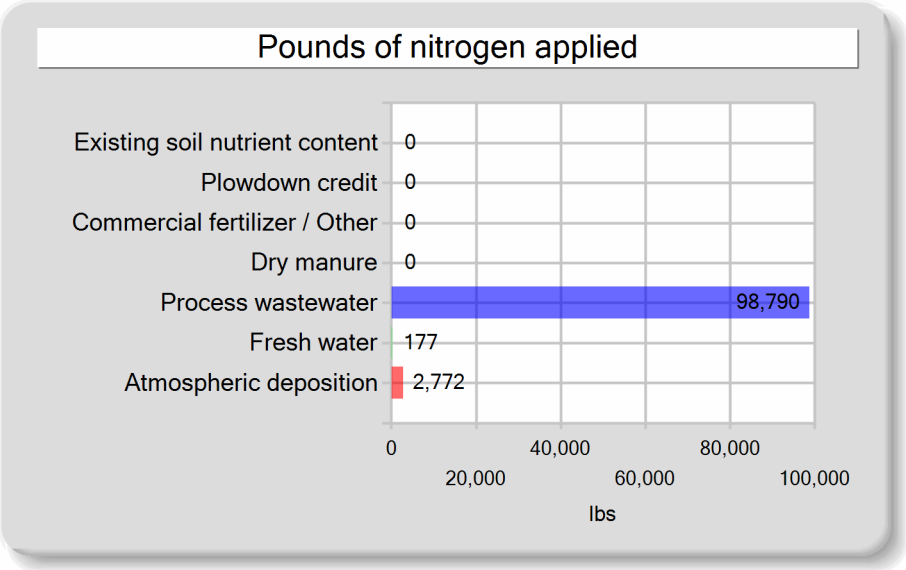
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

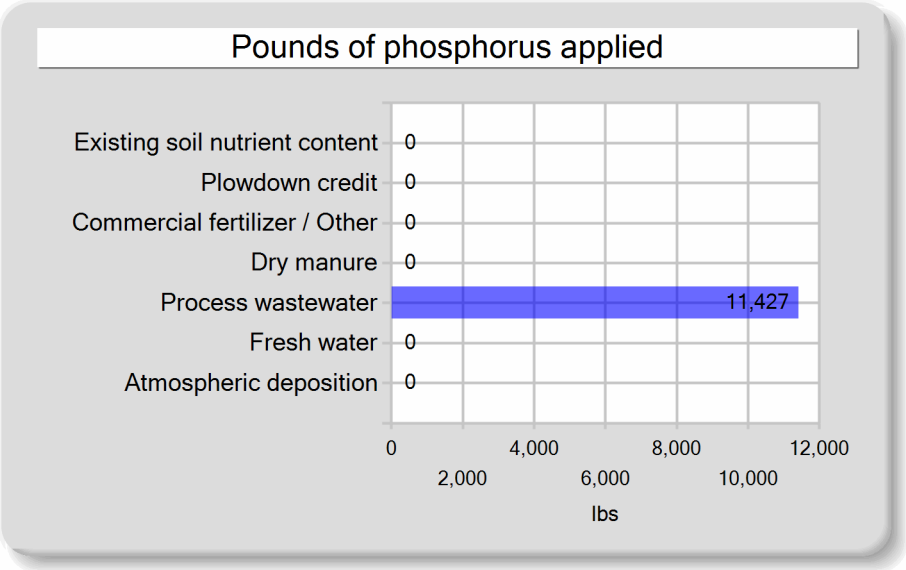
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	98,789.75	11,426.65	171,779.93	964,849.40
Fresh water	176.72	0.00	0.00	56,114.95
Atmospheric deposition	2,772.00	0.00	0.00	0.00
Total nutrients applied	101,738.47	11,426.65	171,779.93	1,020,964.35
Anticipated crop nutrient removal	81,180.00	14,295.60	67,141.80	0.00
Actual crop nutrient removal	84,845.48	21,110.36	148,821.86	550,987.82
Nutrient balance	16,892.99	-9,683.71	22,958.07	469,976.53
Applied to removed ratio	1.20	0.54	1.15	1.85

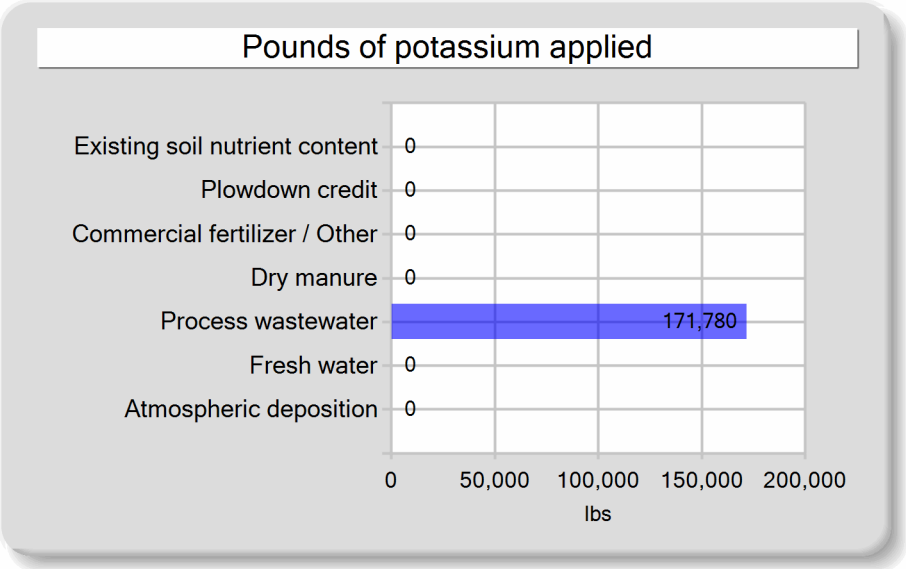
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

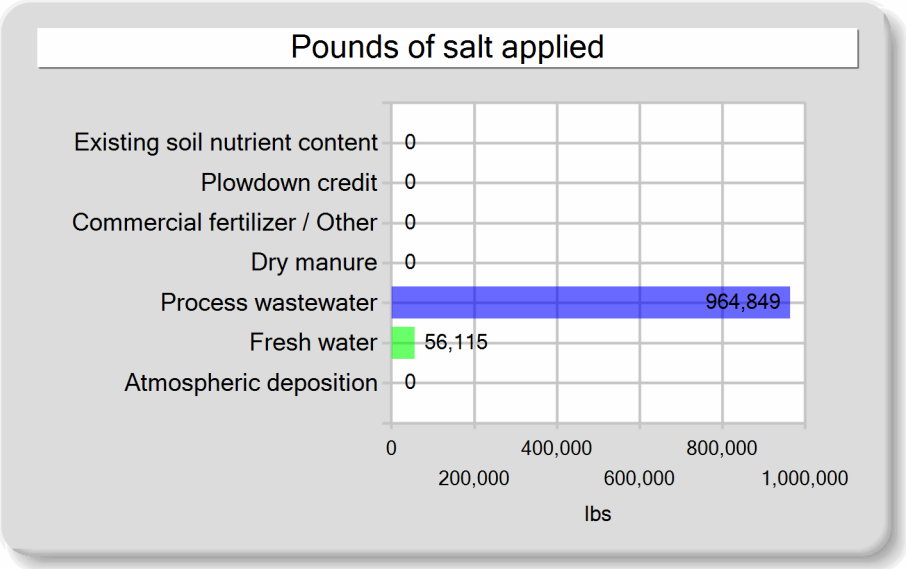


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Jim Hofman (Apr 1, 2024 16:59 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Jim Hofman	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Apr 1, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Jim Hofman

Name of Dairy Facility: Gerrit Hofman Dairy

Facility Address: 4820 Perrin Road, Manteca, CA 95337

Number and Street | City | Zip Code

Contact Person Name and Phone Number: Jim Hofman 209-612-3064

Name | Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Gerrit Hofman Dairy

Address of Hauling Company /Person: 4820 Perrin Road Manteca 95337

Number and Street | City | Zip Code

Contact Person: Jim Hofman 209-612-3064

Name | Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Hofman Brothers 4820 Perrin Road Manteca 95337 209-823-3000

Name | Number and Street | City | Zip Code | Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

29665 Kasson Road Tracy 95337 San Joaquin CO. 255-027-053

Number and Street | City | Zip Code | Assessor's Parcel Number

Dates Hauled: 5/5/23 to 5/10/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 1760 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 55.9% moisture

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

88 loads at 20 ton per load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ (Apr 1, 2024 16:59 PDT) Date: Apr 1, 2024

Hauler's Signature: _____ (Apr 1, 2024 16:59 PDT) Date: Apr 1, 2024

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Jim Hofman

Name of Dairy Facility: Gerrit Hofman Dairy

Facility Address: 4820 Perrin Road, Manteca, CA 95337

Number and Street | City | Zip Code

Contact Person Name and Phone Number: Jim Hofman 209-612-3064

Name | Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Gerrit Hofman Dairy

Address of Hauling Company /Person: 4820 Perrin Road Manteca 95337

Number and Street | City | Zip Code

Contact Person: Jim Hofman 209-612-3064

Name | Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Hofman Brothers 4820 Perrin Road Manteca 95337 209-823-3000

Name | Number and Street | City | Zip Code | Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

29665 Kasson Road Tracy 95337 San Joaquin CO. 255-027-053

Number and Street | City | Zip Code | Assessor's Parcel Number

Dates Hauled: 10/15/23 to 10/31/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 1,720 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 32.6% moisture

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

86 loads at 20 ton per load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ (Apr 1, 2024 16:59 PDT) Date: Apr 1, 2024

Hauler's Signature: _____ (Apr 1, 2024 16:59 PDT) Date: Apr 1, 2024

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Gerrit Hofman
4820 E. Perrin Rd.
Manteca, CA 95337

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/5/23 12:48 PM
SUBMITTED BY: Vince
ANALYZED DATE: 12/6/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
North Domestic Well	T3339014A	Electrical Conductivity (EC)	EPA 120.1 and	0.506	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.176	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	324	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	93.6	98.8 (T3335012B)	101 (T3335012B)	1.80
Electrical Conductivity (EC)	94.5			1.88
Nitrate Nitrogen (NO3-N)	94.9	96.6 (W3339003A)	94.8 (W3339003A)	1.90

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Gerrit Hofman
4820 E. Perrin Rd.
Manteca, CA 95337

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/5/23 12:48 PM
SUBMITTED BY: Vince
ANALYZED DATE: 12/6/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
South Domestic Well	T3339014B	Electrical Conductivity (EC)	EPA 120.1 and	0.520	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.296	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	333	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	93.6	98.8 (T3335012B)	101 (T3335012B)	1.80
Electrical Conductivity (EC)	94.5			1.88
Nitrate Nitrogen (NO3-N)	94.9	96.6 (W3339003A)	94.8 (W3339003A)	1.90

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Compliance Analysis Report

Customer
Gerrit Hofman
4820 E. Perrin Rd.
Manteca, CA 95337

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/5/23 12:48 PM
SUBMITTED BY: Vince
ANALYZED DATE: 12/6/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
West Domestic Well	T3339014C	Electrical Conductivity (EC)	EPA 120.1 and	0.516	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.196	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	330	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	93.6	98.8 (T3335012B)	101 (T3335012B)	1.80
Electrical Conductivity (EC)	94.5			1.88
Nitrate Nitrogen (NO3-N)	94.9	96.6 (W3339003A)	94.8 (W3339003A)	1.90

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

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Compliance Analysis Report

Customer
Gerrit Hofman
4820 E. Perrin Rd.
Manteca, CA 95337

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/5/23 12:48 PM
SUBMITTED BY: Vince
ANALYZED DATE: 12/6/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Silage Pit Domestic Well	T3339014D	Electrical Conductivity (EC)	EPA 120.1 and	1.12	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.78	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	716	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	93.6	98.8 (T3335012B)	101 (T3335012B)	1.80
Electrical Conductivity (EC)	94.5			1.88
Nitrate Nitrogen (NO3-N)	94.9	96.6 (W3339003A)	94.8 (W3339003A)	1.90

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4820 E. Perrin Rd.
Manteca, CA 95337

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/5/23 12:48 PM
SUBMITTED BY: Vince
ANALYZED DATE: 12/6/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Airport Domestic Well	T3339014E	Electrical Conductivity (EC)	EPA 120.1 and	0.528	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	338	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	93.6	98.8 (T3335012B)	101 (T3335012B)	1.80
Electrical Conductivity (EC)	94.5			1.88
Nitrate Nitrogen (NO3-N)	94.9	96.6 (W3339003A)	94.8 (W3339003A)	1.90

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Gerrit Hofman Dairy
Mailing address:
(not facility address) Manteca, CA 95337
Phone#:

Authorized Copy Released To: F1R Ag

Sampled By: Jim Hofman
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	12-5-23	7:40	North Domestic Well		18.6°		✓				
2	12-5-23	7:50	South Domestic Well		19.2'		✓				
3	12-5-23	8:10	West Domestic Well		19.0'		✓				
4	12-5-23	7:15	Silage Pit Domestic Well		18.8'		✓				
5	12-5-23	7:00	Airport Domestic Well		18.6'		✓				
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged In By:	LABORATORY NUMBER
1st	Nina Fontana	F1R Ag	12-5-23 8:30	12-5-23		T3539014
2nd		DENELE ANALYTICAL, INC.	12-5-23 12:48			

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)

2023 Annual Report Gerrit Hoffman

Final Audit Report 2024-04-01

Created:	2024-04-01
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAA1SEY_a-6kwu8woTTKc4h1E5ZSQFQ6Om0

"2023 Annual Report Gerrit Hoffman" History

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